

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; glm-5

Abstract

This report evaluates `z-ai/glm-5` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Observed cost in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie; OpenRouter leads Throughput in all routing modes; Infron leads E2E latency in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie.

Overall, Infron's cross-mode strengths are Streaming TTFT, E2E latency, while OpenRouter's cross-mode strengths are throughput, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput OpenRouter wins 4/4 Max advantage 104.15%, higher is better	Observed cost Infron wins 2/4 Max advantage 62.64%, lower is better	E2E latency Infron wins 3/4 Max advantage 64.76%, lower is better	Streaming TTFT Infron wins 3/4 Max advantage 47.57%, lower is better	Cache hit rate OpenRouter wins 3/4 Max advantage 4.00%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 7.85%	OpenRouter advantage 62.64%	Infron advantage 10.78%	Infron advantage 12.55%	Infron advantage 1.72%
Price First price	OpenRouter advantage 104.15%	OpenRouter advantage 20.72%	OpenRouter advantage 64.76%	OpenRouter advantage 47.57%	OpenRouter advantage 4.00%
Latency First latency	OpenRouter advantage 19.66%	Infron advantage 7.35%	Infron advantage 3.14%	Infron advantage 33.16%	OpenRouter advantage 0.23%
TTFT First ttft	OpenRouter advantage 9.81%	Infron advantage 2.54%	Infron advantage 12.04%	Infron advantage 39.74%	OpenRouter advantage 0.27%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie	Overall metrics and provider drill-down
Performance	OpenRouter leads Throughput in all routing modes; Infron leads E2E latency in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Infron	OpenRouter	OpenRouter	Infron	Infron	Infron leads overall (3/5 metrics)
Price First	Minimize request and token cost	OpenRouter	OpenRouter	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (5/5 metrics)
Latency First	Minimize full–response waiting time	OpenRouter	Infron	OpenRouter	Infron	Infron	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first–token time	OpenRouter	Infron	OpenRouter	Infron	Infron	Infron leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

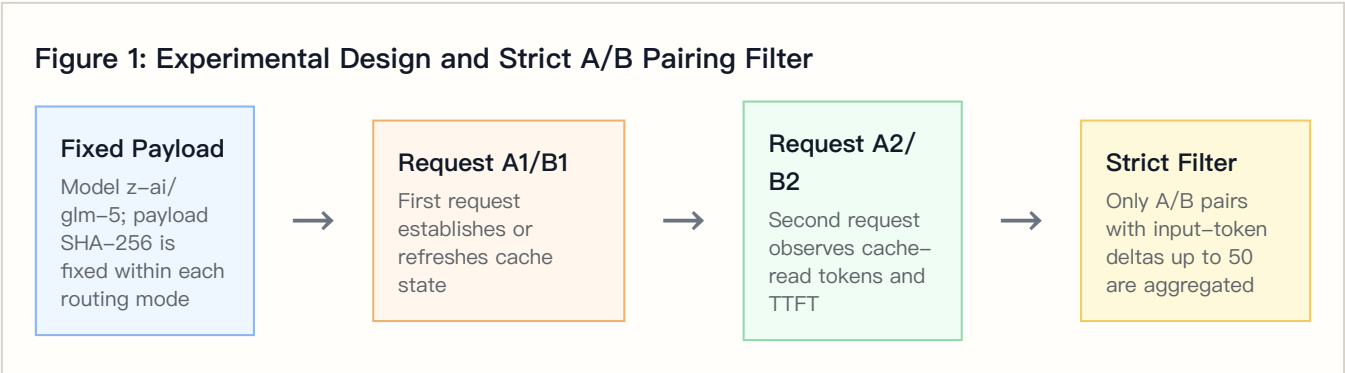
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `business_representative`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	z-ai/glm-5
Provider model IDs	infron: z-ai/glm-5 ; openrouter: z-ai/glm-5
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	24

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	6414682	99.79%	\$1.61315300	2.52 tok/s	5311.71 ms	4992.94 ms
Throughput First	OpenRouter	200	6414682	98.10%	\$0.99183578	2.72 tok/s	5884.51 ms	5619.75 ms
Price First	Infron	195	6040276	95.70%	\$0.99697000	1.52 tok/s	8506.60 ms	7224.73 ms
Price First	OpenRouter	195	6040283	99.53%	\$0.82584095	3.10 tok/s	5163.02 ms	4895.85 ms
Latency First	Infron	193	6273860	99.61%	\$0.77495600	2.61 tok/s	4966.67 ms	3645.69 ms
Latency First	OpenRouter	193	6273860	99.84%	\$0.83192194	3.12 tok/s	5122.85 ms	4854.43 ms
TTFT First	Infron	200	6414682	99.56%	\$0.84264400	2.73 tok/s	4758.14 ms	3596.32 ms
TTFT First	OpenRouter	200	6414682	99.83%	\$0.86408455	3.00 tok/s	5331.20 ms	5025.52 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	5019.43 ms	8949.33 ms	11000.97 ms	4695.00 ms	8587.00 ms	10897.81 ms
Throughput First	OpenRouter	5511.87 ms	9758.45 ms	12789.27 ms	5324.60 ms	9432.64 ms	12480.67 ms
Price First	Infron	4307.77 ms	31280.13 ms	93253.06 ms	3325.72 ms	22342.13 ms	92413.82 ms
Price First	OpenRouter	5101.24 ms	7374.62 ms	8385.21 ms	4839.13 ms	7071.70 ms	7957.55 ms
Latency First	Infron	4716.31 ms	7963.46 ms	9700.74 ms	3452.01 ms	6130.44 ms	8077.04 ms
Latency First	OpenRouter	5123.94 ms	7437.43 ms	8328.80 ms	4865.67 ms	7124.79 ms	8211.71 ms
TTFT First	Infron	4371.01 ms	7859.70 ms	9668.50 ms	3308.84 ms	6277.24 ms	8130.10 ms
TTFT First	OpenRouter	5290.40 ms	7983.31 ms	8954.58 ms	5070.78 ms	7718.23 ms	8577.17 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	1145.60 ms	725.46 ms to 1593.91 ms	<0.001	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	1253.63 ms	821.31 ms to 1704.47 ms	<0.001	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	–0.1442 tok/s	–0.3115 tok/s to 0.0211 tok/s	0.0855	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$–0.00310659	\$–0.00390526 to \$–0.00242740	<0.001	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	1.01 pp	–0.97 pp to 3.01 pp	0.2524	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	–6687.16 ms	–10536.36 ms to –3483.40 ms	<0.001	195	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	–4657.75 ms	–8398.45 ms to –1592.26 ms	0.0065	195	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	–0.6011 tok/s	–0.8364 tok/s to –0.3715 tok/s	<0.001	195	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$–0.00087758	\$–0.00153961 to \$–0.00029056	0.0052	195	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	–4.48 pp	–7.99 pp to –1.34 pp	0.0105	195	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	312.37 ms	–91.12 ms to 681.67 ms	0.1197	193	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	2417.48 ms	2093.52 ms to 2728.71 ms	<0.001	193	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	-0.6766 tok/s	-0.8531 tok/s to -0.5050 tok/s	<0.001	193	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$0.00029516	\$0.00021278 to \$0.00036663	<0.001	193	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	-0.21 pp	-1.29 pp to 0.38 pp	1.0000	193	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	1146.12 ms	764.77 ms to 1497.12 ms	<0.001	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	2858.40 ms	2535.21 ms to 3162.86 ms	<0.001	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	-0.4132 tok/s	-0.5904 tok/s to -0.2386 tok/s	<0.001	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$0.00010720	\$-0.00023742 to \$0.00037466	0.5111	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	-0.62 pp	-2.14 pp to 0.42 pp	0.5049	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	0	0.0000	0	0.00 ms	4992.94 ms	5311.71 ms
Throughput First	OpenRouter	6000	15.0000	400	5619.75 ms	5619.75 ms	5884.51 ms
Price First	Infron	0	0.0000	0	0.00 ms	7224.73 ms	8506.60 ms
Price First	OpenRouter	5835	14.9615	389	4895.85 ms	4895.85 ms	5163.02 ms
Latency First	Infron	0	0.0000	0	0.00 ms	3645.69 ms	4966.67 ms
Latency First	OpenRouter	5955	15.4275	386	4854.43 ms	4854.43 ms	5122.85 ms
TTFT First	Infron	0	0.0000	0	0.00 ms	3596.32 ms	4758.14 ms
TTFT First	OpenRouter	6153	15.3825	400	5025.52 ms	5025.52 ms	5331.20 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	99.62%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	99.44%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3–E2:
Normalized
capability
contour**

Five metrics are normalized to 0–100, where higher is better.

**Figure 3–E3:
Cost–cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure
3–
E1:
Routing–
mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

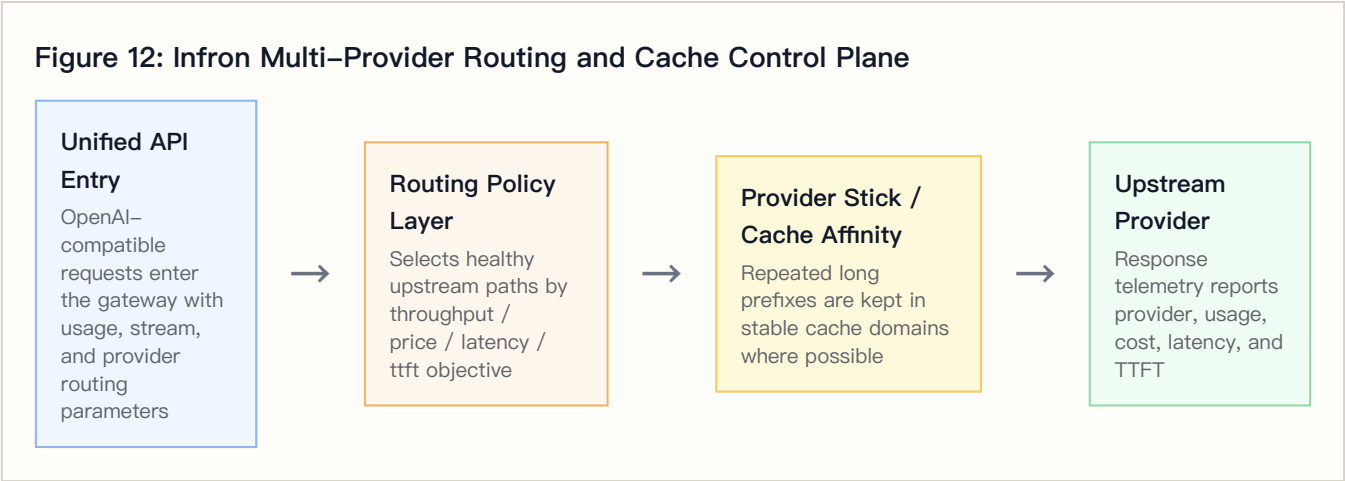
Figure 4–E3: Latency First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	novita 400
Throughput First	OpenRouter	400	400	StreamLake 400
Price First	Infron	390	390	deepinfra 340, alibaba/cn 50

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Price First	OpenRouter	390	390	StreamLake 389, GMICloud 1
Latency First	Infron	386	386	deepinfra 386
Latency First	OpenRouter	386	386	StreamLake 356, DeepInfra 30
TTFT First	Infron	400	400	deepinfra 400
TTFT First	OpenRouter	400	400	StreamLake 362, DeepInfra 31, Baidu 7

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	novita	400	100.00%	200/200	200	4992.94 ms	5311.71 ms	6414682	5357
Throughput First	OpenRouter	StreamLake	400	100.00%	200/200	200	5619.75 ms	5884.51 ms	6414682	6400
Price First	Infron	deepinfra	340	87.18%	165/175	175	3392.45 ms	4571.01 ms	5509975	4395
Price First	Infron	alibaba/cn	50	12.82%	30/20	30	33284.25 ms	35268.63 ms	530301	642
Price First	OpenRouter	StreamLake	389	99.74%	194/195	195	4890.95 ms	5158.59 ms	6002831	6224
Price First	OpenRouter	GMICloud	1	0.26%	1/0	1	6803.91 ms	6883.60 ms	37452	16
Latency First	Infron	deepinfra	386	100.00%	193/193	193	3645.69 ms	4966.67 ms	6273860	5003
Latency First	OpenRouter	StreamLake	356	92.23%	179/177	179	5045.97 ms	5284.98 ms	6114026	5696
Latency First	OpenRouter	DeepInfra	30	7.77%	14/16	16	2581.47 ms	3198.99 ms	159834	480
TTFT First	Infron	deepinfra	400	100.00%	200/200	200	3596.32 ms	4758.14 ms	6414682	5201
TTFT First	OpenRouter	StreamLake	362	90.50%	181/181	182	5283.54 ms	5564.29 ms	6332869	5792
TTFT First	OpenRouter	DeepInfra	31	7.75%	15/16	17	2314.80 ms	2897.23 ms	62022	496
TTFT First	OpenRouter	Baidu	7	1.75%	4/3	5	3686.97 ms	4056.14 ms	19791	112

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache–hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	+1.69 pp	1.63x	novita 100.00%	StreamLake 100.00%	–6000	Infron has higher cache but higher cost; inspect upstream unit price and completion/ reasoning tokens
Price First	–3.83 pp	1.21x	deepinfra 87.18%	StreamLake 99.74%	–5835	OpenRouter has higher cache and lower cost; provider/ cache–domain mix is the main signal
Latency First	–0.23 pp	0.93x	deepinfra 100.00%	StreamLake 92.23%	–5955	Cache and cost move in different directions; evaluate with speed metrics
TTFT First	–0.27 pp	0.98x	deepinfra 100.00%	StreamLake 90.50%	–6153	Cache and cost move in different directions; evaluate with speed metrics

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second–request cache–read tokens, token–level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt–length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	265	466794	451616	96.75%	\$0.17568300	4717.31 ms	3737.61 ms
short	1500	OpenRouter	265	466794	457728	98.06%	\$0.15285428	4144.39 ms	3863.12 ms
medium	8000	Infron	265	2445560	2377760	97.23%	\$0.80691000	5713.62 ms	4668.79 ms
medium	8000	OpenRouter	265	2445560	2414592	98.73%	\$0.71005725	5368.63 ms	5127.16 ms
long	32000	Infron	258	9659396	9579360	99.17%	\$3.24513000	7236.92 ms	6211.62 ms
long	32000	OpenRouter	258	9659396	9613888	99.53%	\$2.65077169	6657.73 ms	6349.72 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	96.63%	98.10%	OpenRouter
short	Price First	93.95%	98.10%	OpenRouter
short	Latency First	98.98%	98.10%	Infron
short	TTFT First	97.53%	97.94%	OpenRouter
medium	Throughput First	99.86%	96.88%	Infron
medium	Price First	92.42%	98.37%	OpenRouter
medium	Latency First	98.30%	99.87%	OpenRouter
medium	TTFT First	98.37%	99.86%	OpenRouter
long	Throughput First	99.92%	98.40%	Infron
long	Price First	96.68%	99.91%	OpenRouter
long	Latency First	99.96%	99.92%	Infron
long	TTFT First	99.96%	99.92%	Infron

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.62%	\$0.62126500	8914.84 ms	8587.00 ms
Infron	2	50	50	99.85%	\$0.33495900	7957.49 ms	7805.95 ms
Infron	3	50	50	99.84%	\$0.33214400	9439.19 ms	9017.67 ms
Infron	4	50	50	99.84%	\$0.32478500	8212.44 ms	8029.86 ms
OpenRouter	1	50	50	99.84%	\$0.30959630	12487.28 ms	12033.90 ms
OpenRouter	2	50	50	98.73%	\$0.22781642	7022.19 ms	6776.78 ms
OpenRouter	3	50	50	94.10%	\$0.24543220	9107.64 ms	8817.53 ms
OpenRouter	4	50	50	99.84%	\$0.20899086	7794.67 ms	7290.17 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	45	45	87.04%	\$0.32151800	87203.15 ms	82998.06 ms
Infron	2	50	50	98.79%	\$0.26056300	10749.09 ms	7560.04 ms
Infron	3	50	50	95.09%	\$0.22211700	7220.94 ms	5011.36 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	4	50	50	99.68%	\$0.19277200	5882.42 ms	4402.77 ms
OpenRouter	1	45	45	99.81%	\$0.17744723	7879.73 ms	7585.10 ms
OpenRouter	2	50	50	98.73%	\$0.22302410	7733.01 ms	7034.32 ms
OpenRouter	3	50	50	99.84%	\$0.21637876	6704.80 ms	6390.40 ms
OpenRouter	4	50	50	99.84%	\$0.20899086	7257.05 ms	7153.45 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	98.73%	\$0.19645000	8331.75 ms	6502.38 ms
Infron	2	45	45	99.91%	\$0.18854500	7858.70 ms	5589.52 ms
Infron	3	49	49	99.91%	\$0.19837800	7133.50 ms	5437.50 ms
Infron	4	49	49	99.90%	\$0.19158300	8337.99 ms	6631.28 ms
OpenRouter	1	50	50	99.84%	\$0.20897198	6611.28 ms	6450.22 ms
OpenRouter	2	45	45	99.86%	\$0.19904936	7447.01 ms	7141.62 ms
OpenRouter	3	49	49	99.84%	\$0.21566118	7305.77 ms	7092.93 ms
OpenRouter	4	49	49	99.84%	\$0.20823942	7455.78 ms	7164.70 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.68%	\$0.19290200	6925.38 ms	4657.72 ms
Infron	2	50	50	98.79%	\$0.22294100	7377.62 ms	5495.93 ms
Infron	3	50	50	99.91%	\$0.23476100	8549.03 ms	7605.03 ms
Infron	4	50	50	99.90%	\$0.19204000	7022.95 ms	5339.84 ms
OpenRouter	1	50	50	99.84%	\$0.20960816	8273.97 ms	8007.52 ms
OpenRouter	2	50	50	99.85%	\$0.22287306	7871.29 ms	7588.82 ms
OpenRouter	3	50	50	99.84%	\$0.21622772	8257.15 ms	7862.65 ms
OpenRouter	4	50	50	99.81%	\$0.21537561	7708.44 ms	7375.08 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	Infron leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	Good for throughput–first tasks; constrain cost and cache separately
Price First	Minimize request and token cost	OpenRouter leads overall (5/5 metrics)	High–frequency templates, support automation, marketing, RAG prefixes	First–token and E2E response are faster; check cache and cost
Latency First	Minimize full–response waiting time	Infron leads overall (3/5 metrics)	Online chat, agent chains, IDE/ writing assistants, realtime tools	Good for throughput–first tasks; constrain cost and cache separately
TTFT First	Minimize streaming first–token time	Infron leads overall (3/5 metrics)	Streaming chat, realtime copilots, first–screen feedback	Good for throughput–first tasks; constrain cost and cache separately

11. Conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Infron	OpenRouter	OpenRouter	Infron	Infron	Infron leads overall (3/5 metrics)
Price First	Minimize request and token cost	OpenRouter	OpenRouter	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (5/5 metrics)
Latency First	Minimize full–response waiting time	OpenRouter	Infron	OpenRouter	Infron	Infron	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first–token time	OpenRouter	Infron	OpenRouter	Infron	Infron	Infron leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short–window stability but not day–level drift	Add soak tests and repeated windows	Scope conclusions to this run

Limitation	Impact	Next step	Current handling
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	export/glm5_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json
Paired dataset	export/glm5_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv
Request-level dataset	export/glm5_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl
Filtered structured records	export/glm5_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json
Excluded-record audit	export/glm5_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records_excluded.json
Test source	tests/test_rerun_routing_sort_cache_cost_ab.py
Benchmark runner source	scripts/rerun_routing_sort_cache_cost_ab.py
HTML report renderer source	scripts/render_glm52_deepseek_style_report.py
Dataset reference	business_representative built-in representative business templates; request-level export is benchmark_requests